

Compound interest



1 Interest is earned on any accumulated interest. The amount of interest earned in any time period changes from one period to the next.

2

a

	Interest(\$)	Balance(\$)
After 0 years	0	9000
After 1 year	450	9450
After 2 years	472.50	9922.50
After 3 years	496.13	10 418.63

b \$1418.63

c \$10 418.63

3

a

i \$3120

ii \$120

iii \$3120

iv \$124.80

v \$129.79

b \$374.59

4

a \$9779.97

b \$5999.33

c \$15 152.93

d \$10 517.80

e \$5314.47

f \$10 399.13

5 \$3376.53

6

a \$1124.86

b \$1411.58

7 \$10 397.09

8 \$6098.38

9 \$857.45

10 \$472.21

11



12 \$7197.96

13 \$7104.33

14
a \$10 442.26 b \$4349.24 c \$14 929.92 d \$8034.12
e \$8597.84 f \$8581.61 g \$2079.54

15 \$1248.81

16 \$9214.38

17
a \$1759.02 b \$1134.22 c \$7092.99 d \$2243.46
e \$510.76 f \$4278.98

18
a \$22 966.11 b \$22 906.30 c \$22 783.86

19
a \$1794.85 b \$1796.70 c \$1.85

20
a \$11 271.78 b \$12 217.59 c \$11 762.64 d Plan 2

21
a \$8579.84 b \$8557.12 c \$8545.64 d \$8537.96
e \$8534.99 f \$8534.23

22 \$2493.25

23
a \$6000 b 8% c quarterly d 7
e \$4446.15



Spreadsheets

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a 1%

b

	A	B	C	D
1	Quarter	Balance at the beginning of quarter	Interest	Balance at the end of quarter
2	1	2000	20	2020
3	2	2020	20.20	2040.20
4	3	2040.20	20.40	2060.60
5	4	2060.60	20.61	2081.21

c 70 quarters

25

a 0.02

b

	A	B	C	D
1	Month	Balance at the beginning of month	Interest	Balance at the end of month
2	January	1000	20	1020
3	February	1020	20.40	1040.40
4	March	1040.40	20.81	1061.21
5	April	1061.21	21.22	1082.43
6	May	1082.43	21.65	1104.08

c \$268.24

26

a 0.02



b

	A	B	C	D
1	Quarter	Balance at the beginning of quarter	Interest	Balance at the end of quarter
2	1	\$5000	\$100	\$5100
3	2	\$5100	\$102.00	\$5202.00
4	3	\$5202.00	\$104.04	\$5306.04
5	4	\$5306.04	\$106.12	\$5412.16

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	A	B	C	D
1	Month	Balance at the beginning of month	Interest	Balance at the end of month
2	July	\$3000	\$30	\$3030
3	August	\$3030	\$30.30	\$3060.30
4	September	\$3060.30	\$30.60	\$3090.90
5	October	\$3090.90	\$30.91	\$3121.81
6	November	\$3121.21	\$31.22	\$3153.03